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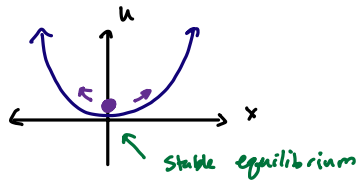
Lecture 21 3/9/26

Summary of last week:

- Hook's Law: $\vec{F} = -k\vec{x} \Rightarrow U(x) = \frac{1}{2}kx^2$



yields SHO "simple harmonic oscillator"

- Any system w/ a stable equilibrium behaves like a SHO near equilibrium (showed using a Taylor expansion)

- EOM for undamped SHO: $\ddot{x} = -\omega^2 x$ $\quad (\omega^2 \equiv k/m)$

$x(t) = B_1 \cos \omega t + B_2 \sin \omega t$ $\qquad B_1 = x_0, \quad B_2 = v_0/\omega$

$x(t) = C_1 e^{i\omega t} + C_2 e^{-i\omega t}$ $\qquad C_1 = \frac{1}{2}(B_1 - iB_2) \quad C_2 = \frac{1}{2}(B_1 + iB_2)$

$x(t) = A\cos(\omega t - \delta)$ $\qquad A =$ amplitude, $\delta =$ phase

$= B_1^2 + B_2^2 \qquad \delta = \tan^{-1}\left(\frac{B_2}{B_1}\right)$

- Damped oscillations: $\vec{F}_{drag} = -b\vec{v}$, $\vec{F}_{spring} = -k\vec{x}$

$$m\ddot{x} = -kx - b\dot{x}$$

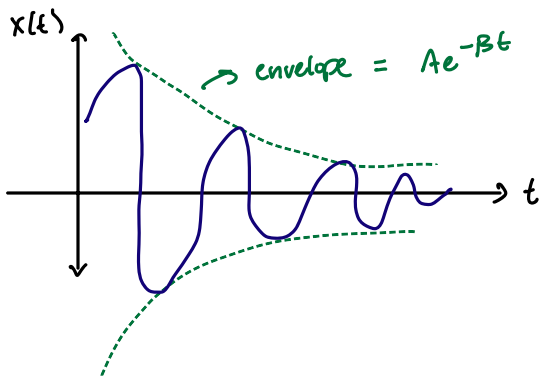
$$\Rightarrow \ddot{x} + 2\beta\dot{x} + \omega_0 x = 0 \qquad (2\beta \equiv b/m, \ \omega_0^2 \equiv k/m)$$

Using $x(t) = e^{rt}$ & plugging into EOM, we found $r_{1,2} = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$

Cases
(I) $\beta = 0$ undamped
(II) $\beta^2 - \omega_0^2 < 0$ underdamped / weak damping
(III) $\beta^2 - \omega_0^2 > 0$ overdamped / strong damping
(IV) $\beta^2 = \omega_0^2$ critical damping

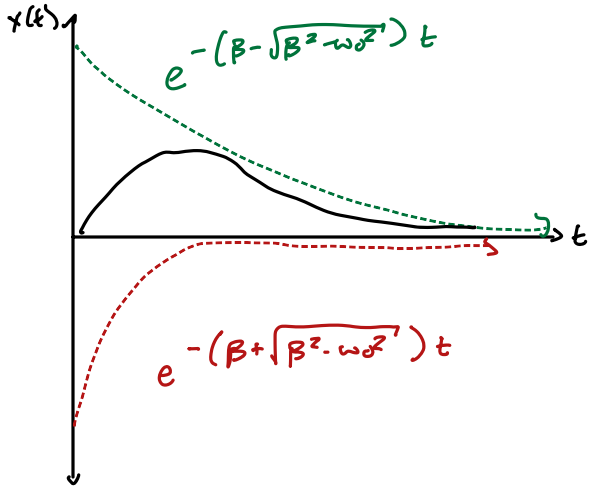
(II) $x(t) = e^{-\beta t}(C_1 e^{i\omega_1 t} + C_2 e^{-i\omega_1 t})$ $\qquad \omega_1 \equiv \sqrt{\omega_0^2 - \beta^2}$

$= Ae^{-\beta t}\cos(\omega_1 t - \delta)$



(III) strong damping

$$x(t) = c_1 e^{-(\beta - \sqrt{\beta^2-\omega_0^2})t} + c_2 e^{-(\beta^2+\sqrt{\beta^2-\omega_0^2})t}$$



IV Critical damping

$\sqrt{\beta^2-\omega_0^2} = 0 \Rightarrow r_1 = r_2$ Problem!! only found <u>one</u> solution

$x_1(t) = e^{-\beta t}$ $x_2(t) = te^{-\beta t}$ → showed in Activity 7

$$x(t) = c_1 e^{-\beta t} + c_2 t e^{-\beta t}$$

Driven Damped oscillations

- Many systems have some external driving force $F(t)$ (time-dependent)

EOM: $\underbrace{m\ddot{x}}_{ma} + \underbrace{b\dot{x}}_{-F_{drag}} + \underbrace{kx}_{-F_{spring}} = \underbrace{F(t)}_{F_{driving}}$

Divide through by m: $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f(t)$, $f(t) \equiv \frac{F(t)}{m}$, $2\beta \equiv b/m$, $\omega_0 \equiv k/m$

<u>Linear Differential operators</u>:

Let $D \equiv \frac{d^2}{dt^2} + 2\beta\frac{d}{dt} + \omega_0^2 \Rightarrow Dx = \ddot{x} + 2\beta\dot{x} + \omega_0^2 x$

The driven damped harmonic oscillator EOM: $Dx = f$

D is <u>linear</u> :

$$D(ax) = a\,Dx$$
$$D(x_1+x_2) = Dx_1 + Dx_2$$

The dampeed harmonic oscillator EOM:

$$Dx = 0$$

we found solutions:

$$\hat{D}x_1 = 0 \quad \Rightarrow \quad \hat{D}(c_1x_1 + c_2x_2) = c_1\hat{D}x_1 + c_2\hat{D}x_2$$
$$\hat{D}x_2 = 0 \qquad\qquad = c_1(0) + c_2(0) = 0+0 = 0$$

superposition!

If x_1 & x_2 are solns, (x_1+x_2) is also a solution

<u>Particular & homogeneous solutions</u>

We want to find x s.t. $Dx = f(t)$

Let's say we make a good guess and find $x_p(t)$:

$$D x_p = f(t)$$

We already have found the solutions to the damped oscillator:

$$D x_H = 0$$

$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$ gave $x(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$ $\Rightarrow$ we call this the <u>homogeneous</u> solution because EOM depends on $x, \dot{x}, \ddot{x}$ only, <u>not</u> directly on t

$$x_H(t) \equiv C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

Now, it turns out that $x_p + x_H$ is <u>also</u> a solution!

$$D(x_p + x_n) = Dx_p + Dx_n = f(t) + 0 = f(t)$$

so we have $Dx_p = f$ $\Rightarrow$ Solution to EOM

$Dx_H = 0$ $\Rightarrow$ <u>not</u> a solution to EOM

$D(x_p + x_H) = f$ $\Rightarrow$ solution to EOM

We just need a single x_p guess & we've solved it!

Sinusoidal Driving Force

Let $f(t) = f_0 \cos(\omega t)$

EOM: $\ddot{x} + 2\beta \dot{x} + \omega_0^2 x = f_0 \cos \omega t$

$$\frac{d^2x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_0^2 x = f_0 \cos\omega t$$

$$\frac{d^2}{dt^2} x(t+\delta) + 2\beta \frac{d}{dt} x(t+\delta) + \omega_0^2 x(t+\delta) = f_0 \cos(\omega(t+\delta)) \qquad \text{Let } \delta = \frac{1}{\omega}\pi/2$$

$$= f_0 \sin\omega t$$

let $x(t+\delta) = y(t)$

$$\frac{d}{dt} x(t+\delta) = \dot{y}$$

$$\frac{d}{dt^2} x(t+\delta) = \ddot{y}$$

$$\Rightarrow \quad \ddot{y} + 2\beta \dot{y} + \omega_0^2 y = f_0 \sin\omega t$$

$$i\ddot{y} + 2\beta i \dot{y} + \omega_0^2 i y = i f_0 \sin\omega t$$

let $z = x + iy$

$$\dot{z} = \dot{x} + i\dot{y}$$

$$\ddot{z} = \ddot{x} + i\ddot{y}$$

$$\Rightarrow \quad \ddot{z} + 2\beta \dot{z} + \omega_0^2 z = f_0 (\cos\omega t + i \sin\omega t) = f_0 e^{i\omega t}$$

$$\ddot{z} + 2\beta \dot{z} + \omega_0^2 z = f_0 e^{i\omega t}$$

$\Rightarrow$ much easier to solve!

different ω's !

Guess the particular soln

$$\left.\begin{aligned} z(t) &= Ce^{i\omega t} \\ \dot{z} &= Ci\omega e^{i\omega t} \\ \ddot{z} &= -C\omega^2 e^{i\omega t} \end{aligned}\right\} \Rightarrow -\omega^2 Ce^{i\omega t} + 2i\beta\omega Ce^{i\omega t} + \omega_0^2 Ce^{i\omega t} = f_0 e^{i\omega t}$$

$$C[-\omega^2 + 2i\beta\omega + \omega_0^2] = f_0$$

$$\boxed{C = \frac{f_0}{\omega_0^2 - \omega^2 + 2i\beta\omega}}$$

$C \in \mathbb{Z}$, and we want to understand this in terms of real quantities:

$C = Ae^{-i\delta}$ I'll leave as an exercise that

$$A^2 = \frac{f_0^2}{(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2}, \qquad \delta = \tan^{-1}\left(\frac{2\beta\omega}{\omega_0^2-\omega^2}\right)$$

So we have

$$z(t) = Ce^{i\omega t} = Ae^{i(\omega t-\delta)} = x(t) + iy(t)$$

$Re(z) = x(t) = A\cos(\omega t-\delta) \equiv x_p(t)$, the particular soln

$$x(t) = x_p(t) + x_h(t)$$

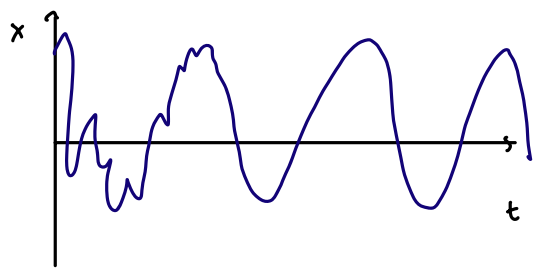
Remember $x_h(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$

$$x(t) = A\cos(\omega t-\delta) + C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

For underdamping, $x_h(t) = Ae^{-\beta t}\cos(\omega_1 t - \delta) \equiv A_{tr} e^{-\beta t}\cos(\omega_1 t - \delta_{tr})$

driven, underdamped: $x(t) = A\cos(\omega t-\delta) + A_{tr} e^{-\beta t}\cos(\omega_1 t - \delta_{tr})$

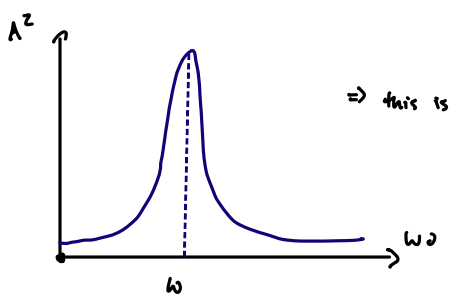
This will lose importance over time



Resonance

$x(t) = A\cos(\omega t-\delta)$ $\qquad A^2 = \dfrac{f_0^2}{(\omega_0-\omega^2)^2 + 4\beta^2\omega^2}$

When the denominator is small, the amplitude is large



$\Rightarrow$ this is called a <u>resonance</u>

denominator $= (\omega_0^2-\omega^2)^2 - 4\beta^2\omega^2$

$\frac{\partial}{\partial\omega_0}$(denominator) $= 2(\omega_0^2-\omega^2)2\omega_0 \Rightarrow$ critical points at $\omega_0 = 0$

$\omega_0 = \omega \rightarrow$ true minimum

$\frac{\partial}{\partial\omega}$(denominator) $= -4\omega(\omega_0^2-\omega^2)^2 - 8\beta^2\omega \Rightarrow$ critical point at $\omega = \sqrt{\omega_0^2-2\beta^2} \equiv \omega_2$